Spatiotemporal Signal Propagation

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1 Part A

1.1 Task A1 — Mutation Comparison

1.1.1 Brief methods summary

To determine if different mutations within the PI3K-AKT pathway alter the strength of spatiotemporal signal propagation, a cohort comparison was performed using the script `compare_spatiotemporal_behavior.py` within the notebook `TaskA1_MutationComparison.ipynb`. The analysis evaluated five cell lines: WT, AKT1_E17K, PIK3CA_E545K, PIK3CA_H1047R, and PTEN_del. Default parameters were maintained across all runs, utilizing a spatial radius of 60 units, a future window of 3 frames, and the ERKKTR_ratio as the signaling column. To establish statistical significance, pairwise comparisons were conducted between the WT control and each mutant line using a two-sided Mann–Whitney U test. P-values were adjusted using the Bonferroni correction method to account for multiple comparisons.

1.1.2 Key results

The group-level summary metrics, standard errors (SE), and corrected statistical outcomes were extracted from `mutations_comparison_table.csv`. The finalized data, highlighting the pairwise comparisons against the Wild Type baseline, is visualized below in Figure 1.

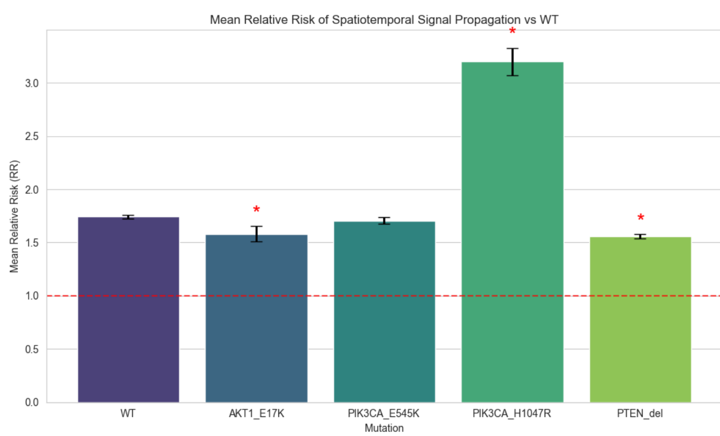

Figure 1: Spatiotemporal comparison: mutants vs WT

1.1.3 Interpretation

The analysis demonstrates that specific mutations within the PI3K-AKT pathway significantly alter the strength of spatiotemporal ERK signal propagation compared to the Wild Type (WT) control (RR=1.74). Following Bonferroni correction, three out of the four mutant lines show statistically significant departures from the baseline, as indicated by the significance markers on the plot. The most prominent effect is driven by the PIK3CA_H1047R mutation, where the mean RR surges to over 3.20. Biologically, this kinase-domain mutation induces constitutive PI3K activity independent of upstream signals, likely leading to an amplified secretion of paracrine factors or enhanced gap junction sensitivity that forces a broader radius of neighboring cells to coordinate their activation.

Conversely, both the AKT1_E17K (RR=1.57) and PTEN_del (RR=1.55) mutations result in a statistically significant decrease in signal propagation compared to WT. This reduction suggests the presence of cell-intrinsic compensatory mechanisms; chronic hyperactivation of downstream AKT via these distinct nodes likely triggers negative feedback loops that partially desensitize receptors to neighboring signals. Meanwhile, the helical-domain PIK3CA_E545K mutation (RR=1.70) does not significantly differ from WT. This divergence highlights that different mutational mechanisms within the exact same signaling pathway can yield fundamentally distinct spatiotemporal behaviors.

1.2 Task A2 — Lagged Exposure

1.2.1 Brief methods summary

To determine if different mutations exhibit distinct spatiotemporal relay timescales, a lagged exposure analysis was performed using the notebook `TaskA2_LaggedExposure.ipynb`. The analysis focused on AKT1_E17K and PTEN_del, as both mutations are known to hyperactivate the AKT pathway through different molecular mechanisms, making them ideal candidates for comparing signaling dynamics. The Relative Risk (RR((τ))) was calculated for lags ranging from (τ)=0 to (τ)=6 frames (0 to 30 minutes). This was achieved by shifting neighbor activation data within individual cell tracks to identify the optimal lag (τ^*) where the signaling influence is maximized.

1.2.2 Key results

The analysis shows that all genotypes reach their maximum signaling influence immediately.

- **Optimal Lag (τ^*):** 0 minutes for all mutations (WT, AKT1_E17K, PTEN_del).
- **Peak RR Values:** PTEN_del (1.57), AKT1_E17K (1.48), and WT (1.34).
- **Persistence:** A major difference was observed in how fast the RR decays. While PTEN_del drops to near-baseline levels rapidly, AKT1_E17K maintains a high RR (1.45) even at the 30-minute mark.

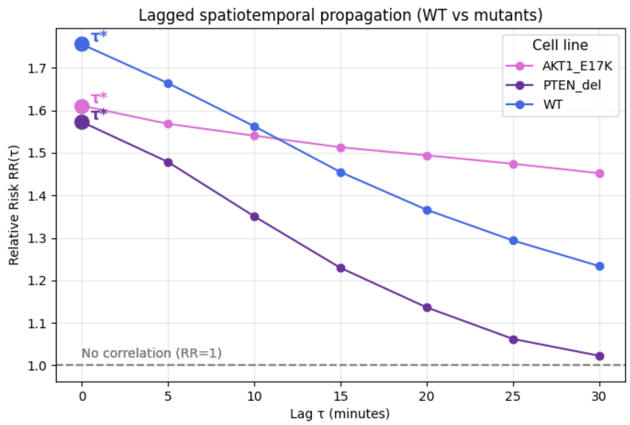

Figure 2: Lagged spatiotemporal propagation

1.2.3 Interpretation

The results show that all three cell lines exhibit an optimal lag of τ=0 minutes*. This implies that the initial propagation of activation from neighbor to focal cell occurs rapidly, within the same 5-minute imaging interval. While this immediate response could theoretically suggest a simultaneous reaction to an external stimulus, the divergence in signal persistence across the mutations indicates a more complex biological communication.

A significant contrast is observed in the signaling duration of the two mutants. In the PTEN_del line, the signaling effect is transient; the RR peaks at 1.57 but drops sharply to near-baseline levels within 30 minutes. This suggests a short-lived "relay" mechanism. In contrast, the AKT1_E17K mutation maintains a much more stable influence, with the RR remaining high (1.45) throughout the entire 30-minute window. Biologically, this suggests that the AKT1 mutation stabilizes the activated state of the signaling pathway or increases the longevity of the secretome. This sustained influence confirms that AKT1_E17K fosters a long-lasting biological communication that is absent or much shorter-lived in the PTEN-deficient line.

1.3 Task A3 — Parameter Robustness Assessment

1.3.1 Brief methods summary

To evaluate the sensitivity of the Relative Risk (RR) metric to analysis parameters, a spatial radius sweep was performed using the notebook `TaskA3_ParameterRobustness.ipynb`. The analysis script was executed multiple times for a fixed experiment-site block (Exp_ID=1, Site_ID=1) while varying the spatial radius parameter (r) across values of 30, 60, 90, and 150 units. All other parameters, such as the future window (3 frames) and jump quantile (0.9), were kept constant. Data was extracted from the resulting summary.json files in the analysis_outputs/ directory to assess how the radius choice influences the detected strength of spatiotemporal coordination.

1.3.2 Key results

The analysis showed that RR decreases as the spatial radius increases. The highest RR was observed at the smallest radius ($r = 30$), while larger radii gradually pushed RR values closer to 1, suggesting that the relationship between neighboring cell activity became weaker.

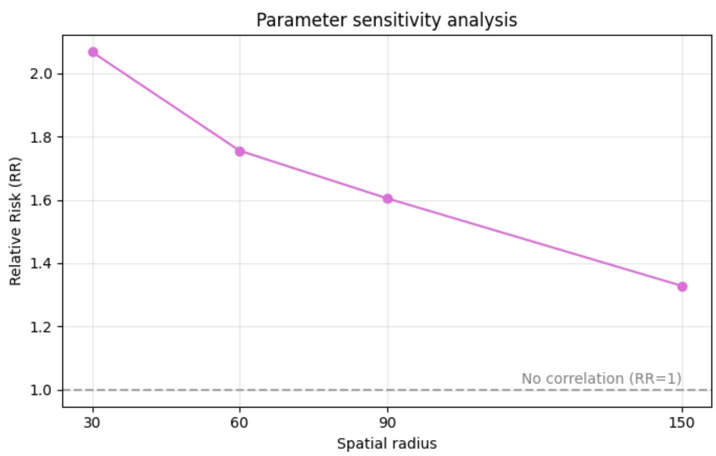

Figure 3: Parameter sensitivity analysis for the spatial radius.

1.3.3 Interpretation

Based on the sensitivity analysis, a spatial radius of 60 seems like the best choice for the downstream analysis.

Although the smallest radius (30) gives the highest Relative Risk (RR = 2.1), it probably reflects interactions between only a small number of neighboring cells, which can make the estimate less stable and more sensitive to noise. On the other hand, increasing the radius to 150 causes the RR to move much closer to 1, suggesting that distant cells add mostly background signal rather than meaningful local interactions. A radius of 60 appears to provide a good compromise: the RR is still relatively strong (1.76), while the number of neighboring cells is large enough to make the measurement more reliable.

From a biological perspective, a radius of 60 units roughly corresponds to a few neighboring cell diameters in epithelial tissue. This is consistent with local communication mechanisms such as paracrine signaling or gap-junction coupling, which are expected to act over short distances rather than across the entire field of view.

2 Independent Research

2.1 Spatial Heterogeneity Analysis

Hypothesis : Cells at the edge of the field-of-view may exhibit different propagation behavior than central cells due to boundary effects or lower neighbor counts.

2.2 Methods

For this task, spatial propagation metrics are calculated for every (experiment id, imaging site pair). No mutation or treatment stratification is applied at this stage. The analysis focuses only on differences between edge and central regions within the field-of-view.

Cell classification The main table was analysed by blocks containing informations from the same experiment and image site. Since a patchy or irregular distribution of the cells was possible, the most perimeter-located nuclei were used as the edge of the field-of-view. Additionally, edges of the preparation were calculated for every frame/time point separately. Distance to the edge was used rather than distance to a microscope image field-of-view centroid.

In order to classify every cell at each time point separately as either centrally or edge-located, for each frame, minima and maxima of both x and y nuclei coordinates were determined. Those boundaries were then treated as borders of the viewed cell group. The edge distance was calculated as a minimum of distances from each cell in that block to the four perimeter boundaries.

Cells with the distance values within the lowest 20% quantile were classified as edge cells, while the remaining cells were classified as centrally located.

Comparison Table For each spatial category, observations are split into two groups: exposed (at least one spatial neighbour is currently jumping) and unexposed (no neighbours jumping). The future jump rate is computed for each group. The relative risk (RR) is then calculated using the formula presented during lectures: $RR = \frac{p_{\text{exp}}}{p_{\text{unexp}}}$, $p_{\text{unexp}} > 0$ Its value above 1 indicates that having an active neighbour increases the probability of a future jump. For statistical analysis, RR is also computed for each individual cell track. A Mann-Whitney-U test is then run comparing the distribution of per-track RRs between edge and central cells. The function returns a summary table (one row per category, containing cell counts, mean neighbor count, jump rates, and RR with its standard deviation) alongside the U statistic and p-value.

For each biosensor separately, the per-site comparison tables were aggregated by spatial category. The standard deviation of RR values was included to capture variability across experiment-site blocks. The final summaries for ERKKTR and FoxO3A were concatenated into a single comparison table. In the end, RR distributions were tested with U-test.

2.3 Results - Comparison Table

cell_category	biosensor	n_cells	mean_neighbours	mean_rr	std_rr
central	ERKKT	153224	11.8535	2.0156	0.7524
edge	ERKKT	61679	8.9622	1.7972	0.596
central	Fox032A	153224	11.8535	1.6774	0.6181
edge	Fox032A	61679	8.9622	1.5012	0.5097

2.4 Results - U-test

- U-test for ERKKT: 5132.0, p-val: 0.0001208
- U-test for FOX03A: 4421.0, p-val: 2e-07

2.5 Interpretation

The standard deviation values indicate moderate variability in RR across different blocks. Nevertheless, central cells consistently exhibited higher mean RR values than edge cells for both biosensors. This suggests that signal propagation events are more strongly coordinated in the central regions of the group of cells.

This may be due to the difference in cell neighborhood structure. Central cells had higher mean neighbor counts than edge cells, increasing the probability of receiving or transmitting propagating signals through local cell-cell interactions.

The Mann–Whitney U-tests confirmed that the RR distributions differed significantly between edge and central cells for both ERKKTR and FoxO3A biosensors. This indicates that the observed differences are unlikely to arise randomly. These findings suggest that spatial context strongly influences propagation dynamics in vitro. In vivo tissues may exhibit different behavior because cells are embedded in more complex three-dimensional environments with additional mechanical and biochemical interactions from outside. Furthermore, cells in vivo located near biopsy boundaries may become mechanically disrupted or partially damaged during tissue preparation, which could additionally alter signaling behavior at the edges.